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Analysis of dynamic bending test using ultra high speed DIC and the virtual fields method



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ABSTRACT

Full-field deformation response of a copper bar impacted in a three point bending setup is studied. Ultra-high speed photography (frame rate: 5×10^6 fps) in conjunction with digital image correlation is utilized to study the in-plane deformation kinematics of the impacted specimen over 25.4 μ s duration after the impact. Distributions of temporal derivatives, i.e. strain rate, velocity and acceleration, over the area of interest are computed using the in-plane displacement and strain fields measured from image correlation. Complex deformation patterns resulting from wave propagation and inertial loading are studied and discussed. Having deformed the specimen well into the plastic regime, the idea initially proposed by Pierron et al. (2011) is extended to identify visco-plastic constitutive response of the examined material by incorporating an analysis based on the method of virtual fields. Accordingly, a strain rate sensitive constitutive model for the copper specimen is identified by reconstructing the inertia and the impact forces using full-field acceleration and reaction force measurements, respectively.

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1. Introduction

Characterizing the mechanical behavior of materials at high strain rates has been a challenge for several decades. Numerous dynamic testing procedures have been devised within the past century to enable the study of the material response when subjected to dynamic loading conditions. Among those, the Kolsky or Split Hopkinson Pressure Bar (SHPB) is one of the most common. The methodology devised in conventional SHPB provides limited information regarding the actual response of the material due to certain restrictions associated with simplifying assumptions, including the use of one-dimensional stress wave propagation [1]. The most significant of these limitations include the applicability of the procedure to uniaxial loading conditions, and the assumption of stress and strain homogeneity in the deforming specimen. The latter is of particular importance in high strain rate deformation of soft and/or low impedance materials, for which delayed state of stress equilibrium arises due to the long stress wave reverberation times [2–5]. In such cases, the non-uniformity of stress within the dynamically deformed specimen can be a direct result of the inertial forces. Unless “inertia effects” are included in the deformation analysis, the accuracy of the results may not be guaranteed, with the level of inaccuracy increasing with increasing strain rate [1]. From an experimental

perspective, direct measurement of inertia forces and their resultant stresses are extremely challenging. However, recent advances in ultra-high speed imaging systems, along with advances in full-field measurement techniques (e.g. digital image correlation) provide a direct way to quantify and include inertia effects in dynamic deformation analyses. In recent years, full-field measurement techniques based on grid method [6–9] and digital image correlation [10–15] have been coupled with high speed imaging equipment to successfully examine the dynamic deformation and failure of various engineering materials. The combination of high speed imaging and full-field measurement techniques facilitates measurement of the entire temporal history of deformations (displacement and strain) at a large number of points within a predefined “area of interest” over a very short course of time. The temporal rate of change of displacement and strain can then be considered for the calculation of time-derivative parameters, i.e. velocity, strain rate and acceleration distributions over the entire field of interest. Among all temporal derivatives, acceleration is of a greater importance, since it can be used to reconstruct the inertia forces [10].

It is well-known that the presence of inertia forces can make the global force measurement insufficient for accurate estimation of the stress history, particularly during the earlier stages of deformation when stress equilibrium is not yet achieved [4,5]. It is during this stage that conventional analyses based on stress equilibrium may not be valid. For stiff engineering materials, e.g. metals, this transient period is very short. Thus, ultra-high speed imaging with very short

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inter-frame times [16] is required since the accuracy and the level of noise in the temporal derivatives are highly sensitive to data frequency. The importance of increased data density when computing temporal derivatives is most readily seen when temporal derivatives are calculated from the displacement and strain fields using numerical algorithms such as finite difference method [10]. Accordingly, inter-frame times on the order of microseconds (or smaller) are necessary in order for an accurate assessment of the acceleration field and the resultant inertia forces. Available ultra-high speed cameras currently permit extremely high frame rates, usually of the order of several million frames per second, but at the cost of spatial resolution and/or temporal duration (number of frames acquired may be quite small). More detailed studies regarding the performance and limitations of a number of commercially available ultra high speed cameras can be found in [10,14,16–19].

The objective in this work is to analyze the full-field deformation response of a bar subjected to dynamic three-point bending. The dynamic three-point bending test is a dynamic testing method widely used to characterize the tensile and flexural strength of quasi-brittle materials [20–22]. In addition, the idea for conducting dynamic three-point bending on notched specimens has also been extensively applied to characterize the dynamic fracture toughness of materials [23–25]. The basic idea in the current study is to take advantage of ultra high speed imaging and full-field measurements to analyze the transient response of a bar subjected to transverse impact. The significance of the present study is that:

(1) Analysis of dynamic deformation is conducted over a time period during which stress equilibrium has not yet been achieved. This is the typical case of a *quasi-impulsive* response, since the test duration is comparable to the time required for the wave propagation in the material. Such conditions commonly occur during dynamic impact testing of brittle materials [20].

(2) The strain magnitudes applied on the copper bar in this work (<5%) are small enough to be typical of the ultimate failure strains of most brittle and quasi-brittle materials under tension and bending [26], while sufficiently large to ensure the presence of plastic deformation and moderate strain hardening in the material. The latter is of greater importance when identification of a constitutive response for the material is to be performed.

The specimen is impacted in a Hopkinson bar apparatus, modified for dynamic three point bend experiments. Full-field deformation response of the material in a 25.4 μ s duration after the impact has been studied via digital image correlation in conjunction with ultrahigh speed photography. Attempts are made to extend the measurement range into the elastic-plastic regime for the copper specimen.¹ Full-field distributions of strain and displacement along with their temporal derivatives (strain rate, velocity and acceleration) are obtained and used to explain the underlying deformation mechanisms in central impact testing of a copper bar. Finally, an analysis based on the virtual fields method (VFM) is performed with the purpose of identifying the visco-plastic constitutive response of the copper specimen.

2. Experimental procedure

2.1. Experimental setup

The experimental setup for the dynamic three-point bending test is shown in Fig. 1a. The setup consists of a custom made three-point bending fixture mounted in a Hopkinson bar apparatus. A 152 mm projectile fired by an air gun is utilized to impact the input bar. The aluminum 7075 input bar is 1822 mm long and 12.7 mm in

diameter. The impactor end of the bar is machined into a U-shaped configuration, as shown in Fig. 1b. The pressure used inside the air gun is 10 psi, and the corresponding velocity of the projectile at the air gun muzzle is ~ 28 m/s. A strain gauge is located 1575 mm away from the tip of the input bar to provide an off-set trigger to the oscilloscope, flash unit and the camera. The triggering time is calculated from the wave velocity of the input bar material ($c_{Al} = 5091$ m/s). Accordingly, a delay time of 309 μ s is calculated, after which the camera begins acquiring images.

As illustrated in Fig. 1b, two piezoelectric load-cells are inserted behind the supports of the three-point bending fixture to measure the reaction force histories. These forces are recorded at 5 MHz by the same oscilloscope used with the strain gauge. The load-cells used in this work are 22.4 kN capacity PCB Piezotronics® load-cells with 224.8 mV/kN sensitivity. These load-cells are designed primarily to measure compressive and impact forces. A polyimide film tap covering the cap surface of the load-cell reduces high frequency ringing associated with metal-to-metal impact.

The specimen is a 10 mm thick copper 110 alloy with 80×6 mm² (L $\times$ W) side dimensions, extracted from the as-received plate. The span between the three-point bend supports is 40 mm, as schematically shown in Fig. 2a. The camera view of the specimen is illustrated in Fig. 2b.

2.2. Ultra-high speed imaging and DIC

A single Shimadzu Hyper Vision HPV-X ultra-high speed camera is used to record images of the deforming specimen. Detailed description of this camera is beyond the scope of this work, but can be found elsewhere [6,14]. A frame rate of 5×10^6 fps is used in this work, resulting in a 0.2 μ s interframe time. The HPV-X camera allows recording of 128 frames with 400×250 pixel² resolution at the utilized frame rate. The exposure time at 5 Million fps is 110 ns. A strobe (Photogenic PowerLight 2500 DR) along with additional high intensity white LED light source is used for illumination. The utilized strobe takes 200 μ s to reach its highest intensity. The light intensity then remains constant for 1 ms before decreasing. Therefore, triggering of the camera and strobe unit must be performed carefully to utilize the maximum possible illumination without significant variation in lighting. This type of illumination provides sufficient light to acquire images up to 5 million frames per second. Details regarding the quality assessment of the utilized strobe unit can be found in [18]. Images are acquired from the speckled side surface of the specimen during the first 25.4 μ s after the impact. Note that experimental impact pulse duration has been approximated as 60 μ s. The 25.4 μ s record time facilitates image acquisition over approximately 40% of the first impact pulse duration. This means that the impactor remains in contact with the specimen over the entire duration of the image acquisition. The record time used in this work is long enough to provide early-stage data that can be used to determine various properties of interest, as detailed in forthcoming sections.

Image correlation process is performed on the area of interest using the software Vic-2D [27]. Note that the speckled area of interest in this work (see Fig. 2b) covers 66 and 250 pixels in x- and y-directions, respectively. Full-field in-plane displacement and strain distributions are computed in this software. Temporal derivatives – velocity, acceleration and strain rate – are calculated separately using the DIC displacement and strain results by applying a central difference scheme:

$$v_i(t) = \frac{u_i(t + \Delta t) - u_i(t - \Delta t)}{2\Delta t} \quad (1)$$

$$a_i(t) = \frac{u_i(t + \Delta t) - 2u_i(t) + u_i(t - \Delta t)}{\Delta t^2} \quad (2)$$

¹ The present study extends the work of Pierron et al. [10], conducted on aluminum bars and using IMACON 200 cameras, where the deformations measured were within the elastic regime due to the limited number of images available.

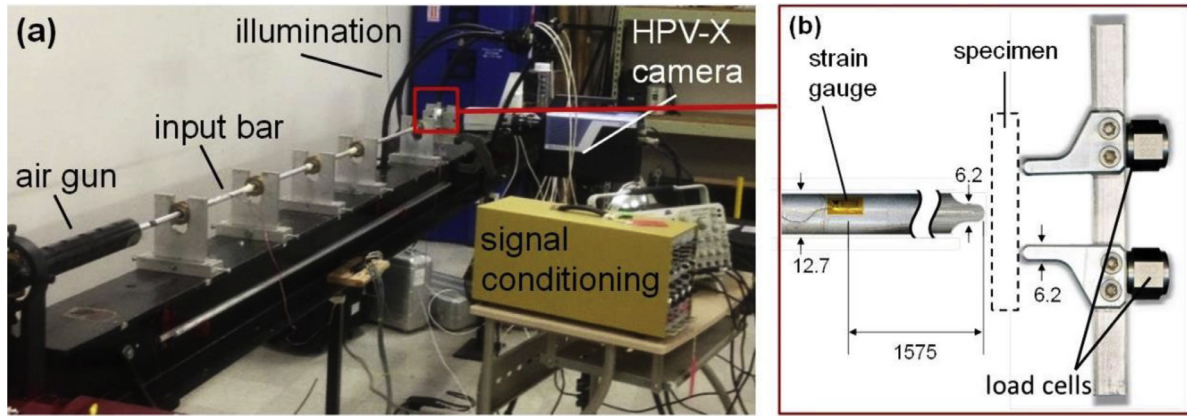


Fig. 1. (a) Experimental setup used for dynamic three-point bending experiment. The three-point bending fixture is shown in (b).

$$\dot{\varepsilon}_{ij}(t) = \frac{\varepsilon_{ij}(t + \Delta t) - \varepsilon_{ij}(t - \Delta t)}{2\Delta t} \quad (3)$$

where u_i , v_i and a_i denote in-plane components of displacement, velocity and acceleration vectors, respectively; and $\dot{\varepsilon}_{ij}$ denotes in-plane strain rate. In these equations, Δt represents interframe time ($=0.2 \mu\text{s}$).

A correlation window (subset) of $23 \times 23 \text{ pixel}^2$, with a 6 pixel step size between subset centers is selected for the image correlation process. Subset dimensions were selected based on a rule of thumb stating that the subset size should be roughly 3–5 times larger than the size of an average speckle [28]. The magnification factor is approximately 11 pixels/mm. Image correlation is performed using the 'normalized squared differences' matching criterion, with an optimized 4-tap interpolation and linear (affine) shape functions. In Vic-2D, strains are derived from the full-field displacement using Gaussian filtering with a filter size of 5 data points, i.e. the smallest filter size allowed in this software. No further spatial data smoothing is performed in this software; though temporal smoothing of the extracted data is conducted prior to the calculation of velocity, acceleration and strain rate fields. Temporal smoothing is conducted using second-order polynomial over $5.5 \mu\text{s}$ time spans. The decision on the use of $5.5 \mu\text{s}$ time span was made based on the results of an extensive sensitivity analysis, details of which are beyond the scope of this work.

3. Results and discussion

3.1. Full-Field displacement and strain distributions

Typical full-field distributions of displacement components in x - and y -directions are shown in Fig. 3. The contour maps depict the displacement fields at $5 \mu\text{s}$, $15 \mu\text{s}$ and $25 \mu\text{s}$ after the impact. Displacement maps in Fig. 3 exhibit patterns that are characteristic of three-point bending experiments [29].

Full-field strain maps are shown in Fig. 4. Fig. 4a exhibits localized compressive ε_{xx} values over an area near to the impact point. Local strain values of up to 3.6% are developed in this location, confirming the occurrence of plastic deformation in the bar. ε_{yy} contour maps in Fig. 4b represent bending strain, with negative (compressive) values on the impact surface, and positive (tensile) values developed on the rear face. Note that the occurrence of plastic deformation in the vicinity of the impacted area causes local stretch on the impact point, resulting in ε_{yy} to take positive values over this region. Maximum tensile strain values of up to 1.8% on the free surface of the bar also confirm the presence of plastic deformation at the back side of the specimen, as well. Full-field shear strain, ε_{xy} , exhibits a typical pattern of shear deformation in three-point bend experiments, with negative and positive values at positive and negative y , respectively. As discussed in [10,30], full-field distribution of shear strains in a dynamic three-point bending experiment can be utilized to reconstruct the bending force applied on the specimen

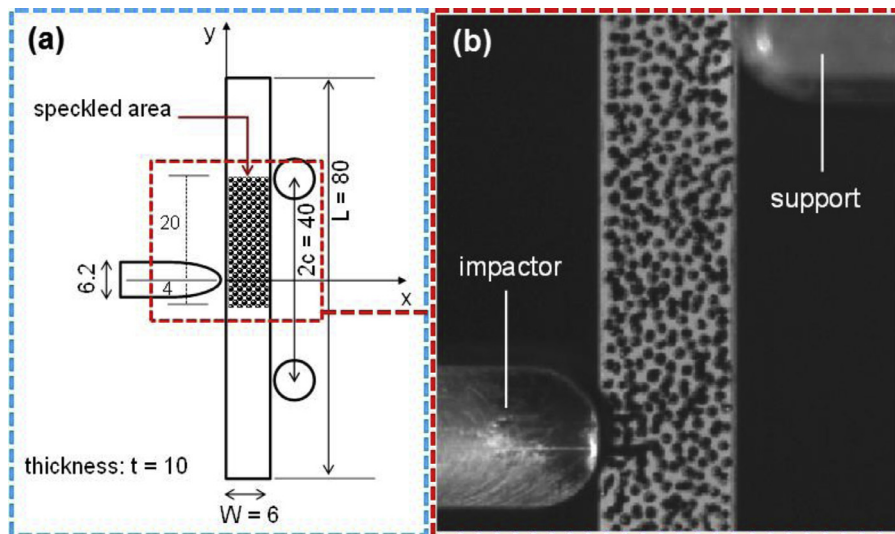


Fig. 2. (a) Schematic of the test specimen showing the location of speckled area of interest. (b) Camera view of the specimen. All dimensions in mm.

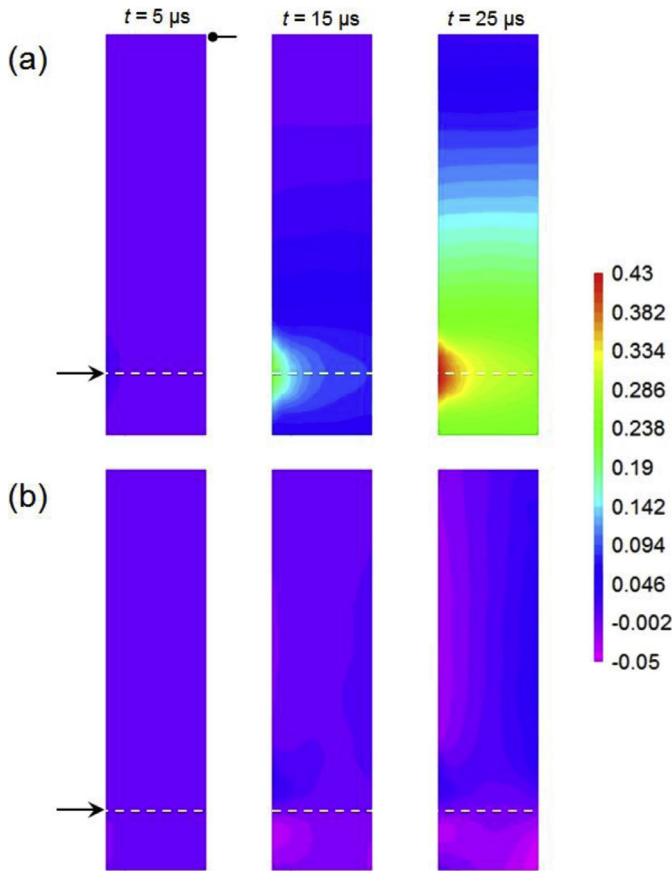


Fig. 3. Distribution of (a) u_x and (b) u_y over the entire field of view, at selected times after the impact. Locations of impact and support are marked. Values are in mm.

using virtual fields method. This requires the use of elastic shear strains and the selection of an appropriate virtual displacement field. A detailed discussion on this subject is out the scope of the present work, but can be found in [10,30].

To identify the visco-plastic constitutive response of the material, total equivalent strain developed in the material is determined. Von Mises equivalent strain, ε_{eq} , developed in the entire field of view can be computed as:

$$\varepsilon_{eq}(x, y, t) = \left[\frac{2}{3} (\varepsilon_{xx}^2(x, y, t) + \varepsilon_{yy}^2(x, y, t) + 2\varepsilon_{xy}^2(x, y, t)) \right]^{1/2} \quad (4)$$

Contour maps showing the evolution of ε_{eq} at different times are shown in Fig. 5. It is clearly evident that ε_{eq} is accumulated very rapidly within areas located closer to the impacted side of the specimen, growing progressively across the entire field of view. Shown in Fig. 6, temporal evolution of equivalent strain averaged over a $5.7 \times 5 \text{ mm}^2$ area of interest indicates total strain buildup to $\approx 2\%$. Note that the smaller region considered here is a subset of the actual area of interest which is more closely associated with the impacted region of the bar, and is symmetric about the x-axis. An interesting point to note in Fig. 6 is that equivalent strain is shown to have built up in two successive stages. An initial increase in the value of equivalent strain is first observed during the first $15 \mu\text{s}$ after the impact. This is followed by a plateau extended over approximately $4 \mu\text{s}$ duration and a second stage of strain accumulation. This behavior is similar to that observed by Pierron et al. [10], and is due to the reversion of the stress wave along the x-axis in the bar. Note that as the stress wave reflects from the free surface of the bar, it opposes the impact force applied on the impacted side of the specimen. At this point, the net force applied on the specimen is at its minimum, thus negligible plastic strain accumulation is occurred in the bar. Strain buildup

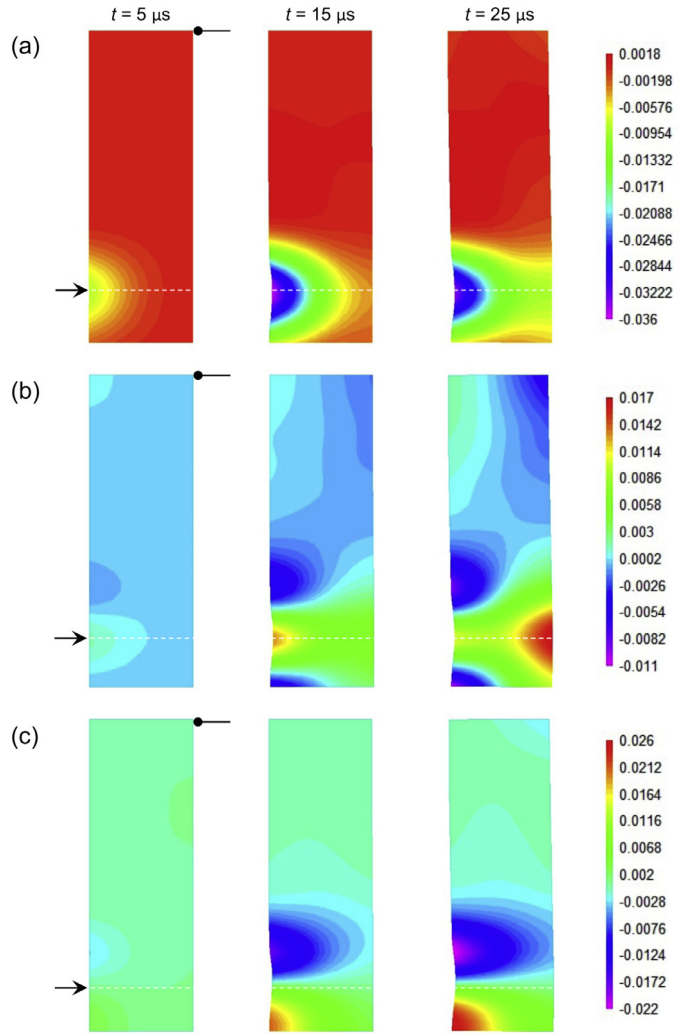


Fig. 4. Distribution of (a) ε_{xx} , (b) ε_{yy} and (c) ε_{xy} over the entire field of view, at selected times after the impact. Locations of impact and support are marked. The arrow indicates the location of impact ($y = 0$).

continues as the stress wave again reflects from the impacted side. More discussions on this are provided in Section 3.3.

3.2. Full-field distributions of velocity, strain rate and acceleration

Distribution of temporal derivatives, i.e. velocity, strain rate and acceleration are considered next. Typical velocity maps determined from smoothed displacement data are shown in Fig. 7. Contour maps show noticeable differences both in pattern and the values of horizontal and vertical velocity components. The horizontal component of velocity, v_x , basically takes values one order of magnitude higher than v_y , as expected in a three point bending test configuration. An interesting point to notice is the spatial variation of v_x along x-direction. To give better insight regarding this, the velocity curves extracted from points A and B in Fig. 8a, along with the loading tip velocity, have been plotted in Fig. 8b. Loading tip velocity in this work is calculated by tracking the tip area of the impactor in time and performing (a) DIC to extract the horizontal displacement vs. time and (b) numerical time differentiation (see Eq. (1)) with smoothing to compute the impactor velocity. Results indicate that there is a time shift of $\sim 5 \mu\text{s}$ between the onset of velocity increase between point A and point B. This time shift is roughly equal to three consecutive reverberation of the elastic stress wave in the x-direction for the copper specimen. After this elapsed time, the

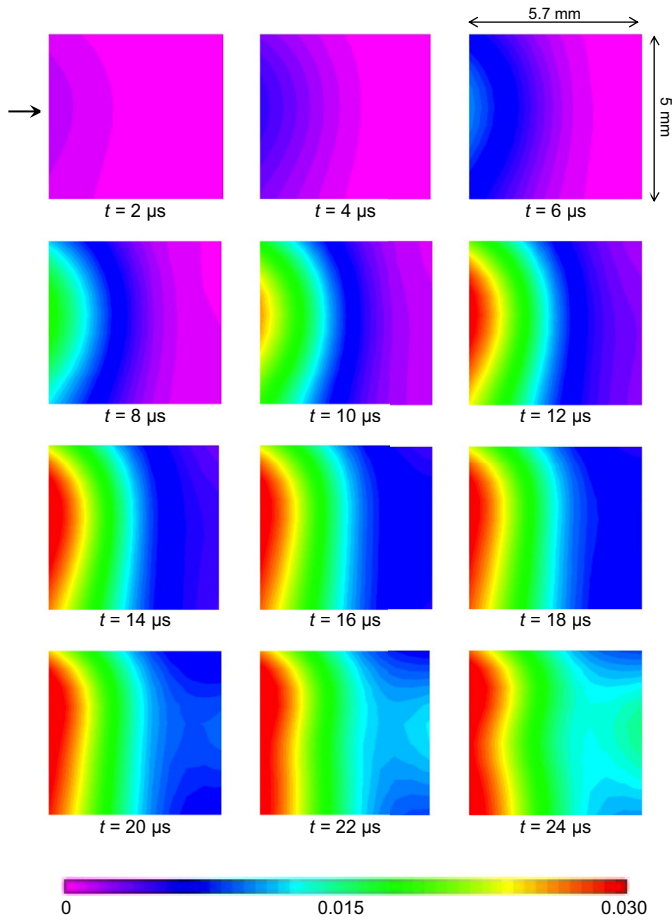


Fig. 5. Evolution of equivalent strain with time. The location of impact ($y = 0$) is marked with an arrow.

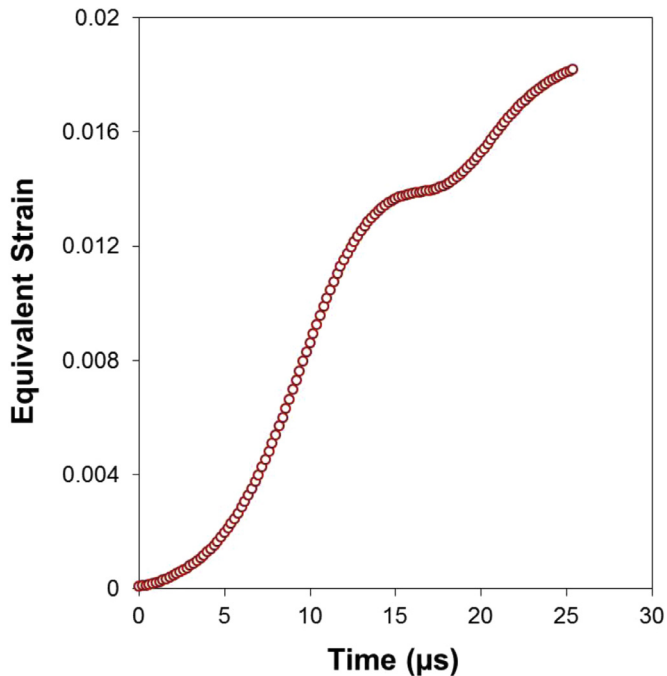


Fig. 6. Temporal evolution of equivalent strain average over the $5.7 \times 5 \text{ mm}^2$ area shown in Fig. 5.

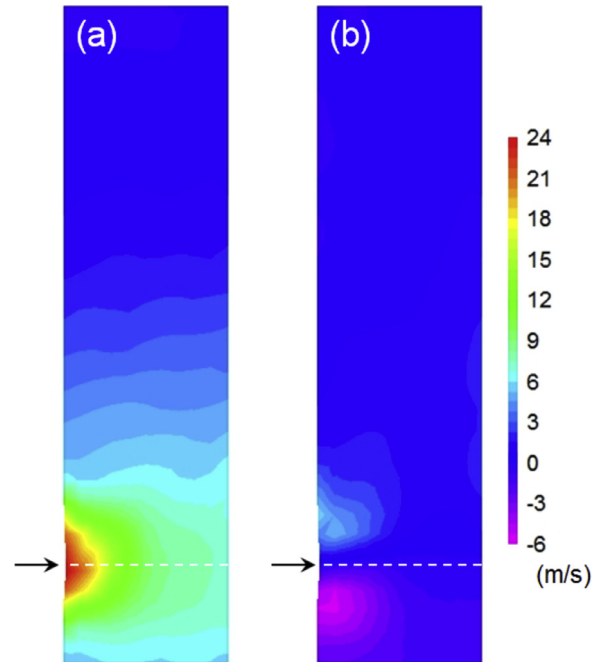


Fig. 7. Distribution of (a) v_x and (b) v_y at $t = 10 \mu\text{s}$. Impactor trajectory is marked with dashed line.

curve obtained for Point B also begins to rise for approximately $10 \mu\text{s}$. Both curves reach a local velocity peak $10 \mu\text{s}$ after the first increase, with similar magnitude for the peak velocity at point A and the impactor tip, as expected. The maximum v_x value for point A at its first peak velocity is about 10 m/s higher than that of point B, which indicates a high spatial variability of v_x . Such substantial variation in v_x gives rise to a highly heterogeneous distribution of strain rate, as discussed later in this section. The velocity peaks are followed by a decrease of 12 m/s and 2 m/s for point A and point B, respectively. This velocity decrease is attributed to the reflection of elastic stress wave from the supports, causing partial deceleration of the impactor tip.

At the point where all local velocity curves reach their local minima ($t = 16 \mu\text{s}$), the velocity within the field of view is expected to be distributed more homogeneously than any other time during deformation. This is investigated by spatially averaging v_x over the $5.7 \times 5 \text{ mm}^2$ area (see Fig. 5) and plotting the average v_x with respect to time, as illustrated in Fig. 8c. The error bars in Fig. 8c denote the standard deviation within the area of interest, thus a measure for the level of spatial variability in v_x over this area. As noted earlier, the lowest spatial variability is found at $t = 16 \mu\text{s}$, i.e. when the reflected wave reaches the impacted side and decelerates the impactor.

Typical full-field distributions of strain rate components at selected times after the impact are shown in Fig. 9. Both $\dot{\epsilon}_{xx}$ and $\dot{\epsilon}_{yy}$ show highly nonhomogeneous distributions at $t = 5 \mu\text{s}$ and $t = 10 \mu\text{s}$; whereas the degree of spatial strain rate variability is reduced at $t = 16 \mu\text{s}$. This, as discussed above, is related to the temporary homogenization of the velocity fields as the result of impactor deceleration.

As illustrated in Fig. 9b, local bending strain rate $\dot{\epsilon}_{yy}$ ranges up to 2000 s^{-1} locally. To determine the average strain rate applied on the specimen in this work, equivalent strain rate, $\dot{\epsilon}_{eq}$, is computed over the area of interest, using an equation similar to Eq. (4), as:

$$\dot{\epsilon}_{eq}(x, y, t) = \left[\frac{2}{3} \left(\dot{\epsilon}_{xx}^2(x, y, t) + \dot{\epsilon}_{yy}^2(x, y, t) + 2\dot{\epsilon}_{xy}^2(x, y, t) \right) \right]^{1/2} \quad (5)$$

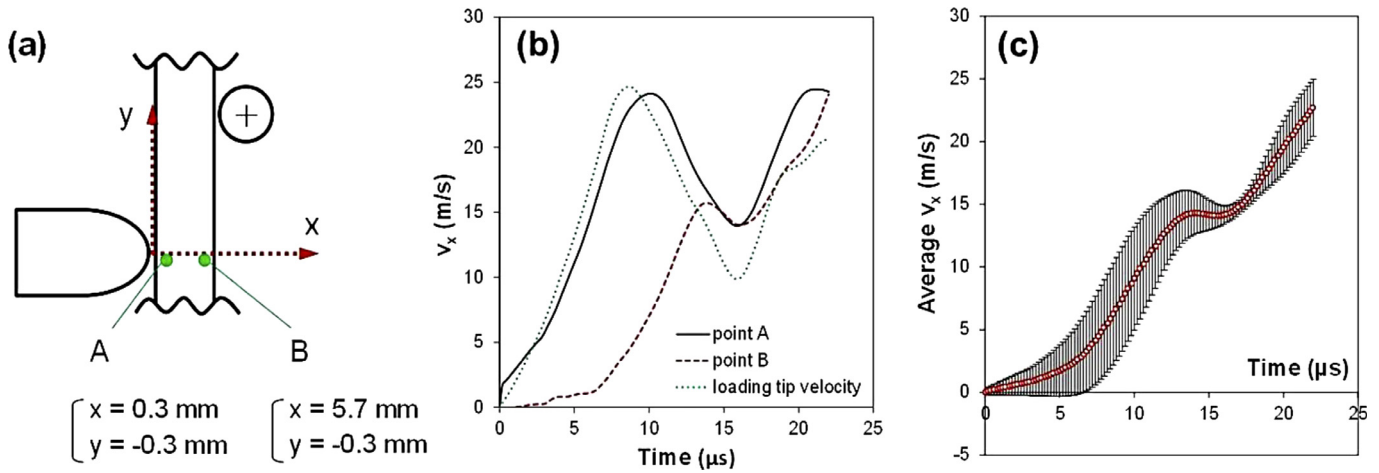


Fig. 8. (a) Location of two representative points from which velocity curves are extracted. (b) Variation of the impactor velocity and v_x for points A and B. (c) Evolution of spatially averaged v_x over the $5.7 \times 5 \text{ mm}^2$ area of interest shown earlier in Fig. 5.

Fig. 10 illustrates the temporal variation of equivalent strain rate averaged over the $5.7 \times 5 \text{ mm}^2$ subset area of interest close to the location of impact. Average equivalent strain rate shows an initial increasing stage up to $t = 10 \text{ μs}$, at which average strain rate reaches 1800 s^{-1} . A remarkable inhomogeneous distribution of strain rate is present in the field of interest at this point of time (see Fig. 9). The curve then follows a decreasing trend, reaching a local minimum at $t = 16 \text{ μs}$, i.e. the time of uniform velocity distribution, as shown earlier in Fig. 8.

The significantly high spatial and temporal variability of strain rates in this work indicates that the material being tested experiences a wide range of strain rates during its course of deformation. Therefore, it can be of great interest to attempt to retrieve strain rate-dependent constitutive laws from a single experiment such as the one conducted in this work. This has been previously achieved by the use of virtual fields method [30–33]. Accordingly, one particular objective in the present work has also been

towards the identification of a visco-plastic constitutive model for the examined material, as presented and discussed in forthcoming sections.

Full-field distribution of horizontal, a_x , and vertical, a_y , acceleration components are shown in Figs. 11 and 12, respectively. After the first establishment of the impact, a localized region containing large a_x values is formed at the impacted side of the specimen, on which acceleration reaches up to $4 \times 10^6 \text{ m/s}^2$. To better understand the temporal variation of acceleration, average a_x and a_y values have been plotted in Fig. 13. The spatial averaging in this figure has been conducted over the entire area of interest. Notice that temporal variation of a_x shows a slight irregularity during the initial 2.75 μs time. This is due to the absence of neighboring information for the displacement data located at the end-points. As a result, temporal smoothing of the displacement data within the first and last 2.75 μs (half width of the smoothing span) may not result in the same level of accuracy as it does on the rest of the data.

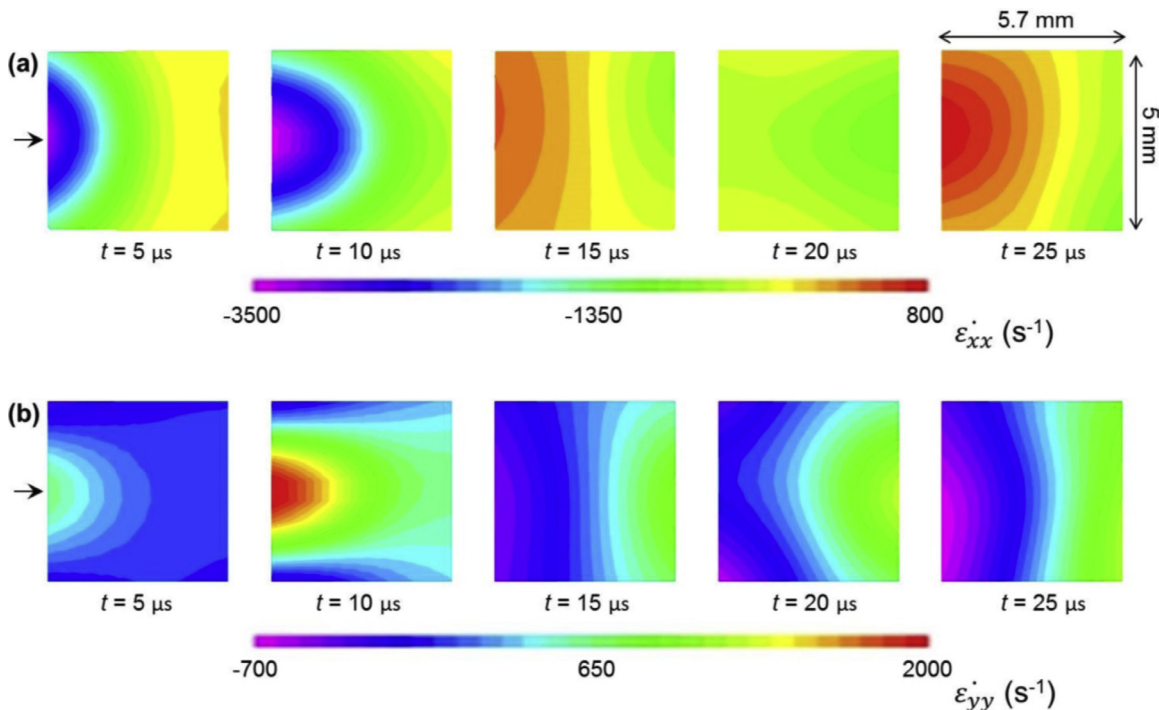


Fig. 9. Distribution of (a) $\dot{\epsilon}_{xx}$ and (b) $\dot{\epsilon}_{yy}$ at selected times after the impact.

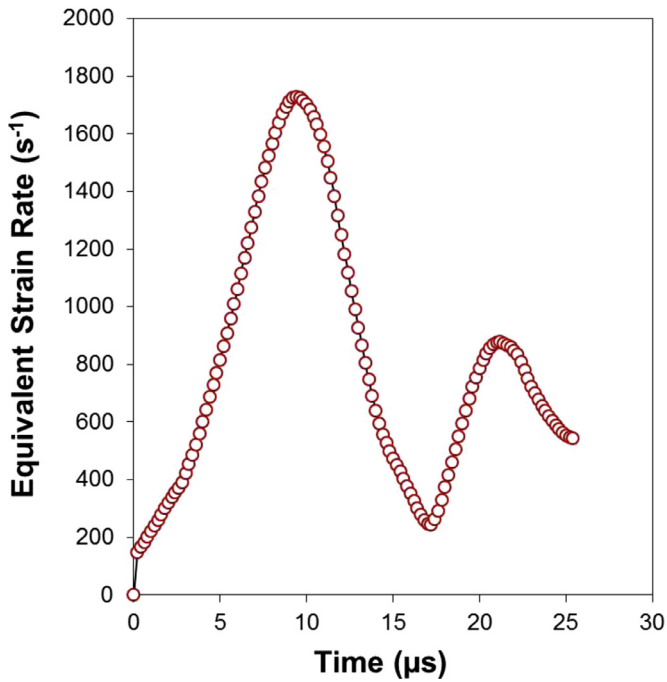


Fig. 10. Temporal evolution of equivalent strain rate average over the $5.7 \times 5 \text{ mm}^2$ area shown in Fig. 6.

Average a_x values are one order of magnitude greater than a_y . It is also evident that a_x follows a periodic trend with a period of $\sim 10 \mu\text{s}$. The $10 \mu\text{s}$ period of oscillation in a_x is equivalent to the time of roughly 3 elastic wave traverses along the width of the copper bar. Considering the longitudinal wave velocity of the copper specimen examined in this work ($c_{Cu} = 3500 \text{ m/s}$), the time for one complete wave reverberation can be calculated as $3.5 \mu\text{s}$. The reason why there is no particular deceleration during the wave reverberation time in this work is that the impactor remains in contact with the

specimen over the entire duration of the analysis and thus, the force exerted by the impactor opposes the resultant force of deceleration.

3.3. Determination of impact force

The impact force can be estimated using the reaction forces measured by the load-cells placed behind the supports of the three-point bend fixture, superimposed with the average inertia force developed inside the dynamically deformed specimen. This idea has previously been implemented in reconstruction of impact force from the boundary-measured and inertia loads [3,11,14]. In the specific case of dynamic bending, the reaction force is measured by two load-cells inserted behind the supports in the three-point bend fixture (see Fig. 1). Fig. 14 shows the reaction forces measured by the two load cells. The total reaction force is determined as the sum of the forces measured by the load-cells. There is an evidence of $<20\%$ difference between the maximum forces measured by the load-cells, which might be due to a slight misalignment of the loading tip and/or the copper bar within the bending fixture. Note that our calculations are based on the total reaction force, i.e. the sum of force histories recorded by the two load-cells. Thus, the final results will be unaffected by the slight differences in the reaction forces measured by the load-cells.

It is observed that the stress wave reaches the load-cells with a $9.2 \mu\text{s}$ delay after the impact. This delay is due to the elapsed time between the instant at which the impact is first established and the time when the stress wave reaches the load-cells. It is realized that the load-cells are not capable of recording the impact load during this “transient” state, while the deformation of the material has already started. Therefore, the use of only reaction forces in the analysis will not assure a realistic characterization of the deformation response of the material.

However, the directly-measured reaction forces can be used in conjunction with inertial loads to facilitate determining the actual impact force applied on the impacted side of the specimen [11]. Accordingly, one can simply estimate the impact force, F , by superimposing the average inertia force with the reaction force measured

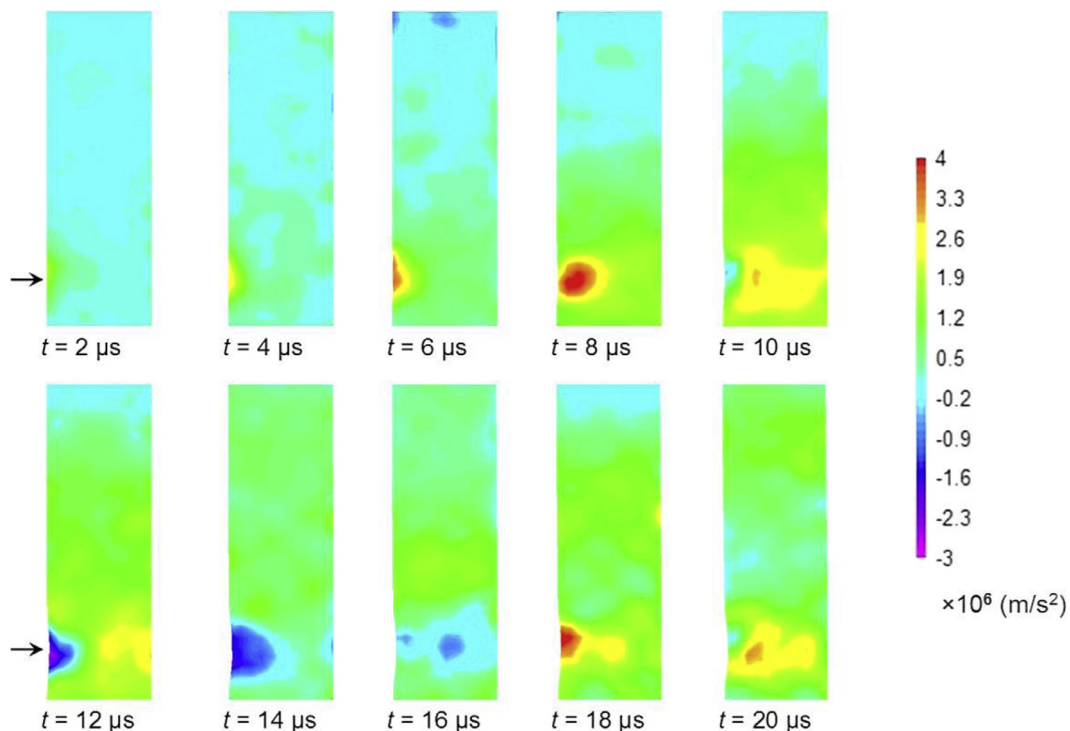


Fig. 11. Distribution of a_x over the entire field of view at different times after the impact. The acceleration values are computed from the smoothed displacement.

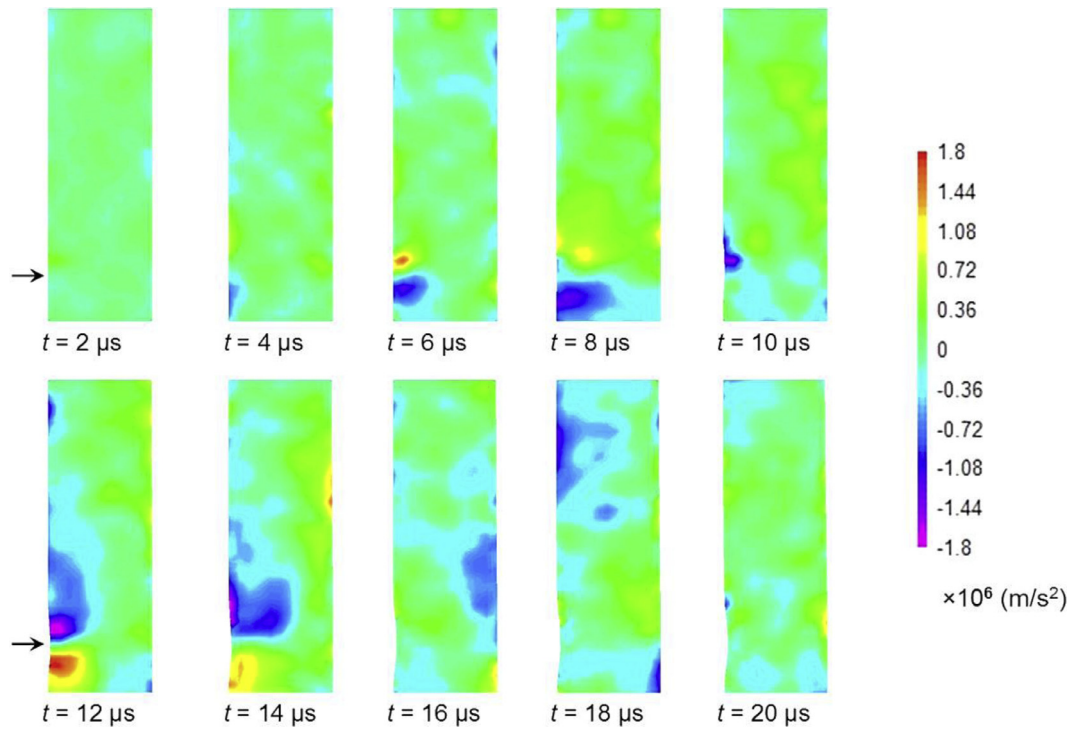


Fig. 12. Distribution of a_y over the entire field of view at different times after the impact. The acceleration values are computed from the smoothed displacement.

at the back-side of the specimen. Average value of inertia force in x -direction can be evaluated from the average acceleration multiplied by mass of the portion of the bar confined between the two supports (see Fig. 2) [10]. The impact force applied on the impacted side of the bar is therefore calculated as:

$$F \cong F_r + (2ctw\rho)\bar{a}_x \quad (6)$$

where F_r is the reaction force measured by the load-cells, and $\bar{a}_x$ is the average horizontal acceleration of the specimen. Neglecting the contribution of the specimen ends located away from the two supports, $\bar{a}_x$ can be evaluated by spatially averaging the

entire population of acceleration data measured on the specimen surface. Evolution of impact force estimated using the reaction force and the average inertial force is plotted in Fig. 15. Impact force increases monotonically during the first $12 \mu s$ after the impact. During this time, a large portion of the impact force is consumed in accelerating the specimen, rather than deforming it. After this period, the total impact force drops due to the reversion of the stress wave in the bar and the consequent reduction in acceleration. The impact force reaches a local minimum at $t = 15 \mu s$, and continues to build up again upon the second reverberation of the stress wave.

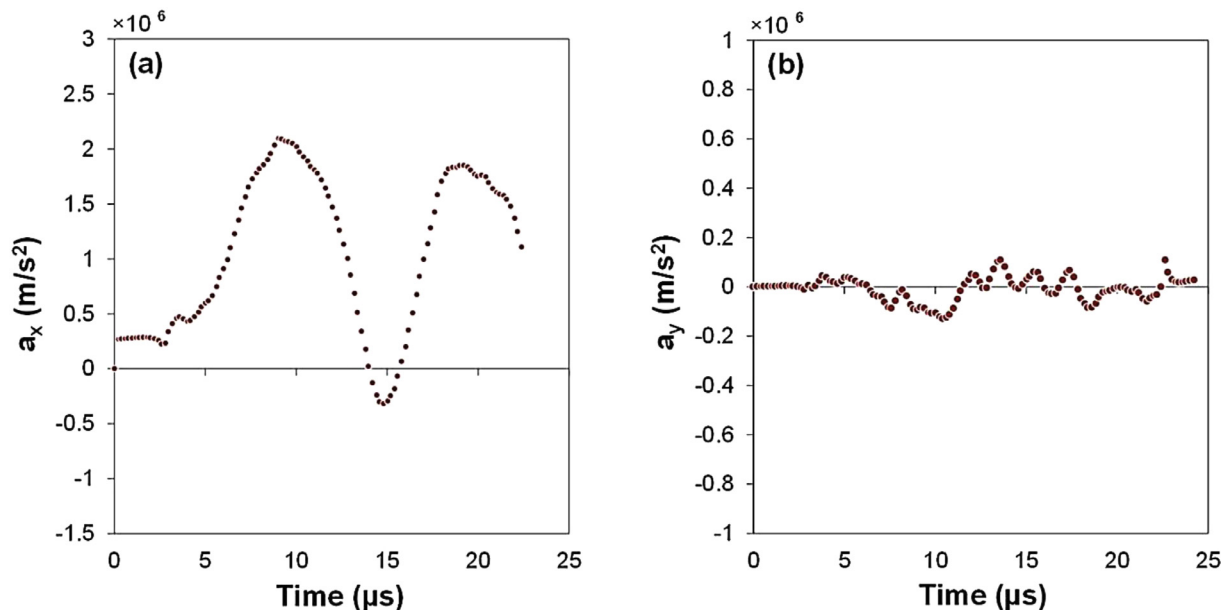


Fig. 13. Temporal variation of (a) a_x and (b) a_y .

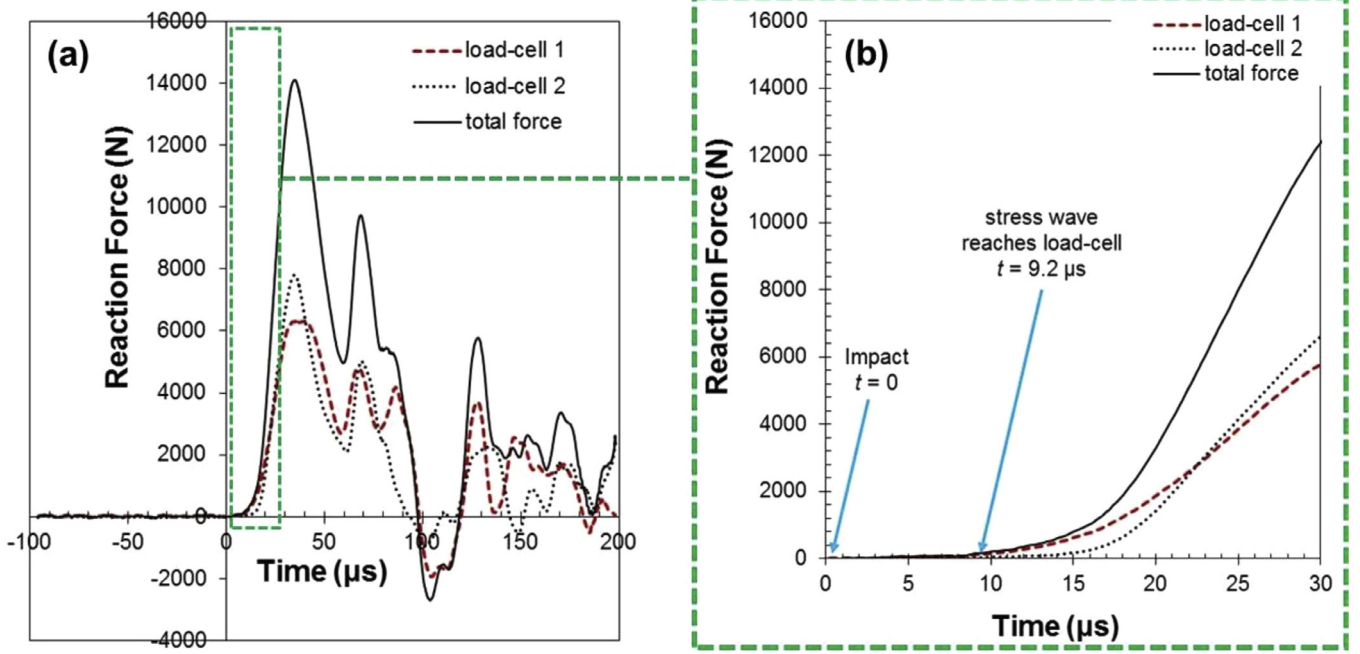


Fig. 14. (a) Temporal variation of reaction force measured by the load-cells. A close-up view of the reaction force history during the record time is shown in (b).

Note that the accuracy of data presented in Fig. 15 over the initial and final 2.75 μs durations is not verified due to the presence of smoothing error discussed earlier in Section 3.1, therefore the VFM analysis is conducted over the middle 20 μs of the record time.

3.4. Visco-plastic constitutive model parameter identification

An analysis based on the method of virtual fields is performed in this work to identify the visco-plastic constitutive response of the copper specimen over the applied ranges of strain and strain rates.

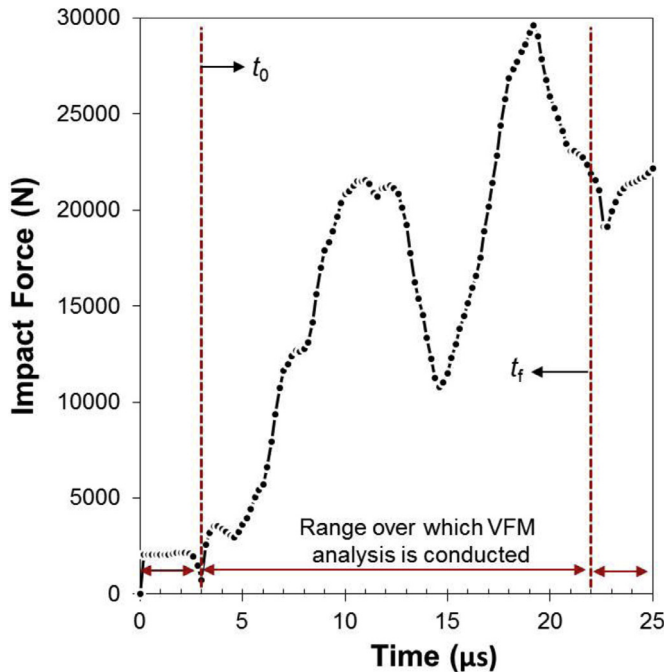


Fig. 15. Variation of bending force in time. t_0 and t_f denote the extents over which VFM analysis is conducted.

For this purpose, the analysis conducted by Pierron et al. [10] is adopted. The principle of virtual work is expressed as:

$$-\int_V \underline{\underline{\sigma}} : \underline{\underline{\varepsilon}}^* dV + \int_{S_f} \underline{T} \cdot \underline{u}^* dS + \int_V \underline{f} \cdot \underline{u}^* dV = \int_V \rho \underline{a} \cdot \underline{u}^* dV \quad (7)$$

where $\underline{\underline{\sigma}}$ and $\underline{\underline{\varepsilon}}^*$ are stress and virtual strain tensors, respectively. $\underline{T}$ is the traction vector acting on S_f . $\underline{u}^*$ denotes the virtual displacement vector and $\underline{f}$ represents volumetric force vector, the contribution of which is neglected in this work. ρ denotes material density and $\underline{a}$ is the acceleration vector. This principle is valid for any kinematically admissible displacement and strain fields.

The principle of virtual work has been used in conjunction with full-field measurements to successfully identify constitutive properties of materials under quasi-static and dynamic loading conditions. The concurrent application of this principle along with the full-field measurements is referred to as the virtual fields method (VFM) within the experimental mechanics' community [30]. For the specific case of dynamic three-point bending, Pierron et al. [10,30] have suggested a virtual field as:

$$\begin{cases} u_x^* = \frac{1}{2}(c^2 - y^2) \\ u_y^* = xy \end{cases} \quad \begin{cases} \varepsilon_{xx}^* = 0 \\ \varepsilon_{yy}^* = x \\ \varepsilon_{xy}^* = 0 \end{cases} \quad (8)$$

over the area of interest shown in Fig. 16. The volume of the specimen over which the above virtual field is defined includes the entire portion of the bar confined between the two supports, neglecting the contribution of the free ends of the specimens located away from the supports. The assumption is justified by the negligible movement of the two specimen ends compared with the amount of deformation applied on the impacted portion of the bar between the two supports [10]. Note that the virtual displacement field considered here is a Bernoulli-like bending displacement with no virtual shear strain. Using the virtual strain fields detailed in Eq. (8) and considering Eq. (4), virtual equivalent strain can be determined as:

$$\varepsilon_{eq}^* = \sqrt{\frac{2}{3}}x \quad (9)$$

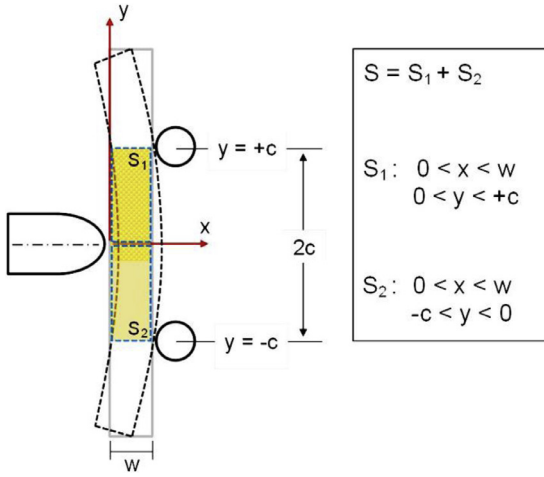


Fig. 16. Details of the bending virtual field.

Assuming that (1) in-plane surface measurements represent the deformation response through the thickness of the bar, and (2) the examined copper is a rigid-plastic material, i.e. contribution of elastic strains is negligible, the first integral on the left hand-side of Eq. (7), i.e. internal virtual work term, can be redefined using the definition of equivalent stress, $\bar{\sigma}$, and equivalent virtual strain, ε_{eq}^* , as [34]:

$$-\int_V \underline{\underline{\sigma}} : \underline{\underline{\varepsilon}}^* dV = -t \int_S \bar{\sigma} \varepsilon_{eq}^* dS \quad (10)$$

where t denotes thickness of the copper bar. Note that based on the quasi-static yield strength of the examined material, the elastic strains are estimated to be less than 0.2%, which can be neglected compared with the total strains of about 2% applied on the specimen in this work. To incorporate the effect of rate sensitivity into the analysis and ultimately identify the visco-plastic constitutive response of the examined material, a simple rate-dependent constitutive model is considered to incorporate the effect of applied strain rate on the flow stress of the material as:

$$\bar{\sigma} = K \bar{\varepsilon}^n \left[1 + C \ln \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}^0} \right) \right] \quad (11)$$

where $\bar{\varepsilon}$ is the equivalent strain and $\dot{\varepsilon}$ is the applied equivalent strain rate. Note that the term $K \bar{\varepsilon}^n$ denotes the constitutive response of the material at reference (quasi-static) conditions. Parameters K and n were determined in this work as 344 MPa and 0.078, respectively, from the quasi-static tension experiments conducted on the same copper specimen at reference strain rate ($\dot{\varepsilon}^0$) of $5 \times 10^{-4} \text{ s}^{-1}$. Strain rate sensitivity of the constitutive response is incorporated by introducing a simple Johnson–Cook type term to the right hand-side of Eq. (11). The objective here is to identify the model parameter C from experimental measurements and the VFM analysis.

Combining Eqs. (10) and (11), and approximating the surface integration with a simple spatial averaging [10, 31], the internal virtual work can be further simplified to:

$$-\int_V \underline{\underline{\sigma}} : \underline{\underline{\varepsilon}}^* dV = -\sqrt{\frac{2}{3}} t w^2 c \left[\left(K \bar{\varepsilon}^n \left[1 + C \ln \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}^0} \right) \right] \right)^{S_1} \right] \quad (12)$$

where the overline indicates the spatial averaging over the area S_1 shown in Fig. 16. Due to symmetry and that the full-field measurements are conducted on the upper half of the entire surface S , Eq. (12) has been manipulated to include the measurement points located only inside the upper half area S_1 , i.e. ($0 < x < w$) and ($0 < y < c$).

At the location of impact ($y=0$), the only nonzero virtual displacement component is $u_x^* = c^2/2$, which is multiplied by the impact

(bending) force, F , to simplify the second integral, i.e. external virtual work, in Eq. (7) as:

$$\int_{S_f} \underline{\underline{T}} \cdot \underline{\underline{u}}^* dS = \frac{F c^2}{2} \quad (13)$$

The integral term on the right hand-side of Eq. (7) denotes the virtual work done by the acceleration and is expanded as:

$$\int_V \rho \underline{\underline{a}} \cdot \underline{\underline{u}}^* dV = t \rho \left[\int_S \frac{1}{2} a_x (c^2 - y^2) dx dy + \int_S a_y x y dx dy \right] \quad (14)$$

which upon simplification yields:

$$\int_V \rho \underline{\underline{a}} \cdot \underline{\underline{u}}^* dV = \frac{\rho t w c^2}{12} \left[8 c \bar{a}_x^{S_1} + 3 w \bar{a}_y^{S_1} \right] \quad (15)$$

Quantities $\bar{a}_x^{S_1}$ and $\bar{a}_y^{S_1}$ in Eq. (15) indicate the horizontal and vertical components of the acceleration spatially averaged over S_1 .

After further rearrangements, Eq. (7) can be re-stated for the specific case of central impact as:

$$F = F_i + F_d \quad (16a)$$

$$F_i = \frac{\rho t w}{6} \left[8 c \bar{a}_x^{S_1} + 3 w \bar{a}_y^{S_1} \right] F_d = \sqrt{\frac{8}{3}} \frac{t w^2}{c} \left[\left(K \bar{\varepsilon}^n \left[1 + C \ln \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}^0} \right) \right] \right)^{S_1} \right] \quad (16b)$$

Eq. (16) states that the impact force, F , in a central impact bending is consumed in accelerating and deforming the specimen. Inertia and deformation forces are denoted in Eq. (16) by F_i and F_d , respectively. Note that the deformation force, F_d , is basically a function of specimen geometry and the local stress data averaged spatially over the area of interest. Having determined the impact force along with full-field acceleration, strain and strain rates, the only unknown in Eq. (16) will remain to be the key model parameter C . The procedure followed here to identify the unknown model parameter C is similar to those documented in [32,33].

In ideal cases – where the selected constitutive model describes the material response correctly, deformation measurements are accurate and noise free, and the virtual field is correctly selected – the difference between the impact force (F) and the sum of deformation and inertia forces ($F_i + F_d$) must be close to zero. In practice, due to the presence of measurement noise and that the selected constitutive model might not be a perfect one for the examined material and/or deformation conditions, the idea will be to construct a cost function, Φ , and minimize it with respect to the sought constitutive model parameters [30]. Accordingly, the cost function in this work is expressed as the summation of the quadratic difference between F and ($F_i + F_d$) as:

$$\Phi(C) = \sum_{t=t_0}^{t_f} \left[F - \frac{\rho t w}{6} \left[8 c \bar{a}_x^{S_1} + 3 w \bar{a}_y^{S_1} \right] - \sqrt{\frac{8}{3}} \frac{t w^2}{c} \left[\left(K \bar{\varepsilon}^n \left[1 + C \ln \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}^0} \right) \right] \right)^{S_1} \right] \right]^2 \quad (17)$$

Lower (t_0) and upper (t_f) bounds of the summation represent the times during which the material undergoes plastic deformation. Note that following the discussion on the invalid acceleration measurements during the initial and final 2.75 μs of the record time, t_0 and t_f are determined as 3 μs and 22 μs , respectively.

Eq. (17) indicates that Φ is an implicit function of local stress values, whereas the local stresses depend on the value of C . Therefore, Eq. (17) is a nonlinear function of the unknown parameter C . Minimization of the cost function, Φ , is achieved in this work through a simple iterative solution. The iterative approach uses an initial estimate for C , such that the local stresses can be determined within the area of interest. The local stresses determined in this way are highly inhomogeneous due to the strong spatial variability of local strain

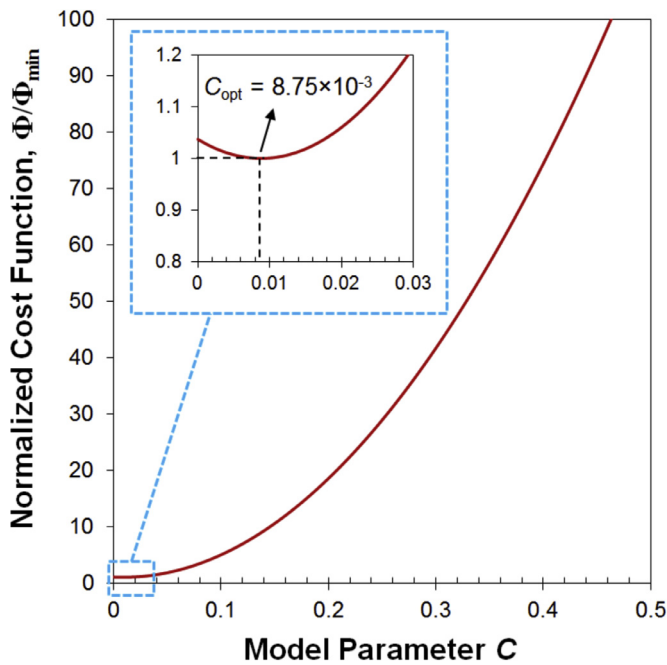


Fig. 17. Variation of normalized cost function $\Phi/\Phi_{\min}$ with model parameter C .

and strain rate data. Local stresses are then spatially averaged and substituted in Eq. (17). Minimization of the cost function $\Phi(C)$ is then carried out for the values of C ranging from 0 to 1 at increments of 2.5×10^{-4} . The value of C that minimizes Φ is taken as the next estimate. The same process is iterated until the C values obtained in two successive steps converge. The final model parameter C obtained after convergence is considered as the strain rate sensitivity parameter for the examined material.

Fig. 17 shows the variation of normalized cost function with the unknown model parameter C obtained after the final convergence check. As a direct result of measurement noise and the over/under-fitting of the constitutive model chosen in the present work, the cost function does not yield a zero value at any C . However, the minimum value for the cost function is achieved at $C_{\text{opt}} = 8.75 \times 10^{-3}$. The optimal value of C obtained in this work is consistent with the values reported for annealed OFHC copper in previous studies, see e.g. [35].

The visco-plastic constitutive model identified here can be further improved and used as a baseline for more complex parameter identification algorithms, e.g. finite element model updating [36,37], to facilitate the acquisition of an elasto-visco-plastic constitutive response for the examined material. The approach described in this work can be used in conjunction with more advanced ultra-high speed cameras with enhanced spatial and temporal resolutions and longer record times to determine the constitutive parameters at larger strains, or possibly using more complex constitutive models. Attempts are currently being made by the authors to implement a similar approach in the study of dynamic bending using full-field measurements conducted over 128 μs ; by performing ultra-high speed imaging at a rate of 10^6 fps, a wealth of experimental data can be obtained. Although the results obtained from such studies might not give information as detailed as those presented in the current work, they can still facilitate a more accurate and more realistic identification of the visco-plastic constitutive law for the material over a wider range of strains and strain rates.

5. Conclusions

A thorough experimental study on the deformation response of three point bending specimen subjected to dynamic loading

conditions was presented. Ultra high speed imaging in conjunction with 2D digital image correlation was implemented to analyze full-field dynamic deformation response of central impacted copper specimen. Ultra high speed imaging was carried out using a single HPV-X camera allowing for the acquisition of 128 images after the impact, with interframe time of 200 ns. The presence and significance of bending waves in the dynamically deformed specimen were evidenced using the full-field acceleration maps, also indicating a very complex deformation response of the impacted specimen in a dynamic three-point bend experiment.

Acceleration fields obtained from the in-plane displacements were used to estimate the average inertia force developed in the dynamically bent copper specimen. Reaction force measured at the location of the three-point bending supports was used in conjunction with the average inertia force to reconstruct the bending force exerted on the impacted side of the bar. The analysis was further extended by implementing an approach based on the Virtual Fields Method, aiming at the identification of the visco-plastic constitutive response of the examined copper specimen. Rate-dependent constitutive response of the material was identified over strain and strain rates ranging from 0 to 0.02, and 0 to 1800 s^{-1} , respectively.

The present work extended a previous study on the concurrent application of ultra-high speed DIC and VFM by providing a detailed discussion on dynamic material behavior of a bar in an impact bending experiment. Taking advantage of a more advanced camera, temporal resolution of the measurement was significantly improved, compared with the original work [10]. Additionally, the challenge due to the number of images acquired during the record time was also resolved by the use of a camera with the capability of recording 128 images upon each trigger. The parameter identification process based on VFM was also pushed further into the plastic regime, where strain rate sensitivity of the material was investigated.

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Corrigendum

Corrigendum to “Analysis of dynamic bending test using ultra high speed DIC and the virtual fields method” [Int. J. Impact Eng. 110 (2017) 299–310]

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ABSTRACT

The authors have identified an inappropriate assumption in the original article that required the authors to provide corrections. They identified an inappropriate assumption in the as-calculated integrals in Eqs. (10) and (14) in the original article [1]. In this regard, integrals denoting internal virtual work (Eq. (10) in the original article) and virtual work of the acceleration (Eq. (14) in the original article) were calculated assuming spatially-constant variables. As shown in the original results, this assumption is inappropriate. Fortunately, updating the assumption only affects results in the last part of the paper, Section 3.4 “visco-plastic constitutive model parameter identification”. Corrections have been made and the final results have been recalculated and presented in this corrigendum. Some additional discussion is also provided to give a deeper insight regarding the enhancement of the constitutive model identification.

1. Determination of virtual work

Fig. 1 represents details of the virtual fields used in this work. An integral expression for the internal virtual work is presented in Eq. (1) [1,2]:

$$-\int_V \sigma: \varepsilon^* dV = \quad (1)$$

wherein the asterisk denotes the virtual strain (displacement) field, adopted from [2]. Using the constitutive law utilized in the original article [1], and assuming that in-plane deformation response measured on the surface is representative of that developed through the thickness, Eq. (1) can be rewritten in the following form:

$$-\int_V \sigma: \varepsilon^* dV = -\sqrt{\frac{2}{3}} t \int_S K \bar{\varepsilon}^n \left[1 + C \ln \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right) \right] x dx dy \quad (2)$$

where t denotes sample thickness. K and n are material constants obtained from quasi-static experiments ($\dot{\varepsilon}_0 = 5 \times 10^{-4} \text{ s}^{-1}$). $\bar{\varepsilon}$ and $\dot{\varepsilon}$ denote equivalent plastic strain and strain rate, respectively. The only unknown in Eq. (1) is the rate-sensitivity parameter C . This parameter is obtained through a minimization algorithm detailed later in this section.

The virtual work done by acceleration can also be expressed as:

$$\int_V \rho \underline{a} \cdot \underline{u}^* dV = \rho \left[\int_S \frac{1}{2} a_x (c^2 - y^2) dx dy + \int_S a_y xy dx dy \right] \quad (3)$$

Determination of the unknown rate sensitivity parameter C in Eq. (2) requires evaluation of the above integrals over the entire area of interest. The area of interest in this work is referred to the surface confined between the two supports in the three-point bend setup (see Fig. 1). These two integrals were evaluated in the original article based on the assumption of spatially-constant strain and acceleration components. This assumption is inappropriate for dynamic bending due to strong heterogeneity of the deformation field, particularly during the transient time, i.e. the period of interest in the present study. Therefore, the integrals have been reevaluated using numerical approaches, taking the spatial variability of in-plane strain and acceleration components into account.

Upon recalculation of internal and inertial virtual work expressions, a cost function is constructed based on the difference between the measured impact force and the sum of internal and inertial forces. This cost function was defined in the original article (Eq. (17)) and will not be repeated here for the sake of brevity. The C value that minimizes the cost function is identified as the rate sensitivity parameter of the material. In this regard, an updated C parameter equal to 1.676×10^{-2} was identified. Accordingly, the final constitutive equation for the copper sample used in this work will be expressed as:

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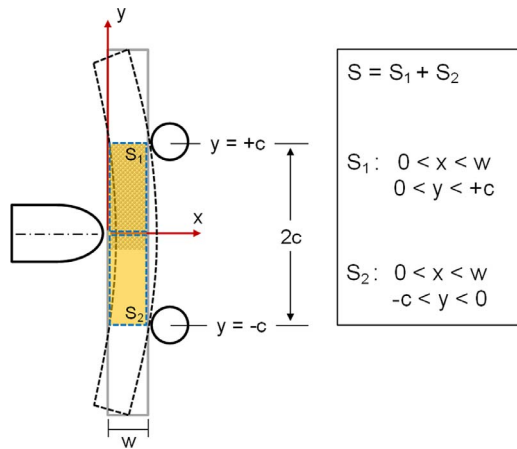


Fig. 1. Details regarding the geometry of the dynamic bending problem analyzed in this work.

$$\bar{\sigma} = 344\bar{\epsilon}^{0.078} \left[1 + 1.676 \times 10^{-2} \ln \left(\frac{\dot{\epsilon}}{5 \times 10^{-4}} \right) \right] \quad (4)$$

2. Further insights

The identification of constitutive response of materials over a wide range of variables using a minimum number of experiments is gaining constantly increasing attention [3–5]. In this regard, the virtual field method (VFM) has been an effective tool. The fundamental idea behind our original article was also to show that such concepts can be extended to include unconventional material testing approaches, such as dynamic bending. There are, however, several details that must be carefully considered to obtain the most accurate results from such experiments. One challenge is that the strain levels achieved are relatively small, resulting in low strain rate ranges. Such small strain and strain rate ranges will carry little weight in the cost function, resulting in poor constitutive model identification.

Another specific challenge in dynamic bending experiments is the sample edge effect. Note that in-plane strain components take the highest values near the edges. However, full-field data obtained from DIC does not include information about deformation response near edges of the sample¹. This is due to the intrinsic nature of DIC algorithm that requires data collection from the center of subsets. This issue has also been addressed in some previous studies by proposing data reconstruction areas that allow for extraction of missing data points near the edges [6]. Finally, results obtained through the approach proposed in this work can be further improved if coupled with accurate finite element (FE) analyses.

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¹ The ability to extend 2D-DIC measurements all the way to the edge of a specimen has been developed recently and is now available for use in VIC-2D commercial software [7].